Environmental fluctuation strengthens gene regulation only when it can be sensed

[**PENDING:** Author names][1*]

1 [**PENDING:** Affiliation]

* [**PENDING:** corresponding author e-mail]

Abstract

Environmental variability is widely expected to favour more elaborate gene regulation, but models disagree even about the direction of the effect, and most of them measure regulatory complexity by counting network connections. We separated fluctuation from the information available for tracking it and asked how each affects regulatory architecture, using two independent individual-based models of genome evolution. In EvoSim the genome is a regulatory network of evolvable size, and in R-Aevol regulation arises from transcription factors encoded in a DNA-like sequence. In each model we compared frozen and fluctuating environments with an informative cue, with no cue, and with a yoked cue that supplies the same signal molecules at the same frequencies but independently of the environment. In EvoSim, fluctuation increased the density of regulatory connections by 26% and reduced wiring strength per possible connection by 11%, so that total wiring strength did not change. In R-Aevol we tested four predictions registered before the simulations were run, at three switching rates and two population sizes, and all four were supported. With an informative cue, fluctuation increased total regulatory strength 1.8- to 2.7-fold relative to a frozen environment, and informative cues produced 2.3 to 2.9 times the regulatory strength of yoked cues. Without a cue, fluctuating populations had the weakest regulation of any condition. Fluctuation also reduced the number of genes by 8 to 10. Cue-free fluctuation had these effects whether an environment lasted half a lifetime or about 250 generations. Connection counts moved in the opposite direction to regulatory strength in EvoSim and in R-Aevol populations with an informative cue. Fluctuation therefore strengthened regulation only when the environment could be tracked, and counting edges can misjudge the direction of the effect.

Author summary

Organisms in changing environments often have complex gene regulation, and it is easy to assume that environmental change is what builds it. We checked this assumption in two computer models of evolving genomes that work in very different ways. Populations evolved in environments that were either constant or switching between two states, and we controlled what the organisms could know about the switching. Some populations received a signal that told them which state they were in. Others received the same signals arranged so that they told them nothing, and a third group received no signal. Switching on its own did not produce stronger regulation. With a meaningful signal, regulatory connections became much stronger than in a constant environment. Without one, they became weaker than in any other condition, whether the environment switched within a lifetime or only every few hundred

generations. The uninformative signal did not help, although it was physically identical to the meaningful one. We also found that counting regulatory connections, a common measure of network complexity, can point the opposite way to measuring how strong the connections are. Our main predictions were written down and committed before the simulations were run.

Introduction

Organisms that live in variable environments are often assumed to need more elaborate gene regulation, and therefore to evolve it. The argument is simple. Regulation lets one genome produce different phenotypes in different conditions, and in a constant environment there is little for it to do. Comparative genomics offers some support. In bacteria the number of transcription factors grows faster than the number of genes [1], roughly as its square [2]. Free-living species carry more transcription factors than their genome size predicts [3], and intracellular parasites and other species of stable niches have simpler signalling systems than species from the open environment [4].

At the level of the phenotype, theory specifies when variability favours a response. A plastic response within a lifetime pays when the environment changes on the timescale of an individual's life and a reliable cue indicates the coming state [5–8]. Without such a cue, fluctuation favours a fixed generalist phenotype, bet-hedging or genetic tracking [7, 8]. A recent model of a gene regulatory network shows the same pattern for the evolution of plasticity. Plasticity evolved mainly under fast, erratic change, especially when cues were reliable, and evolvability under intermediate variability with less reliable cues [9].

Much less is known about the consequences for regulatory architecture, by which we mean the number of genes and the strength with which they regulate one another, and on this point models disagree. Tsuda and Kawata found that fluctuating selection promoted complex regulatory networks, mostly through the fixation of beneficial gene duplications [10]. Their networks received a constant input that carried no information about the moving optimum. Bentkowski and colleagues reported larger genomes in fluctuating environments [11]. Studies with the Aevol platform, on the other hand, find genome size limited by selection for mutational robustness and reduced by large population size and high mutation rates [12, 13]. Work on modularity shows that its evolution depends on sparseness and on how the variation is structured [14, 15].

Differences of method make these results hard to compare. Fluctuation and information are usually confounded. An environment that changes more often also contains a different amount of predictable structure, and adding a cue changes what an organism experiences as well as what it can know. Studies of animal learning handle this problem with a yoked control, in which a second subject receives the same stimuli as the first but without the contingency between stimulus and outcome. Church showed that pairing individual subjects in this way can bias the comparison [16]. In our design each yoked arm receives the same stimulus statistics as the informative arm without being matched to particular informative-arm populations, so this bias does not arise. We know of no use of this design in studies of regulatory network evolution. Measurement is a further problem. Regulatory complexity is usually quantified by counting connections [10, 14, 15], but connection number can grow through the accumulation of nearly neutral interactions [17] and does not reflect how strongly a connection acts. Weighted measures have been used [18], though seldom as the main outcome.

We addressed these issues with two independent models of genome evolution that represent genomes and regulation in different ways. In EvoSim the genome is a regulatory network whose size and connectivity evolve. In R-Aevol, regulation arises from transcription factors encoded in a DNA-like sequence that bind upstream of genes [19, 20]. In each model we compared frozen and fluctuating environments, and under fluctuation we compared an informative cue, no cue, and a yoked cue matched to the informative one in signal identity, frequency and switching rate. Connectivity was measured as a count and as a strength. An exploratory replication in R-Aevol contradicted two conclusions we had drawn earlier from EvoSim. We then stated four predictions, committed them before running new simulations, and tested them at three switching rates and two population sizes. A further pre-registered experiment varied environmental grain to examine the disagreement with Tsuda and Kawata.

Fluctuation strengthened regulation when a cue made the environment trackable and weakened it when no cue was available. The difference came from the information the cue carried, since a cue that carried none had no such effect. Fluctuation also reduced genome size. Without a cue, these effects were the same whether the environment changed within a lifetime or every few hundred generations, so environmental grain does not explain the disagreement with Tsuda and Kawata. Connection counts and connection strength sometimes changed in opposite directions, so a count alone can misrepresent the effect of fluctuation on a network.

Results

Models and experimental design

EvoSim is an individual-based model of unicellular organisms on a two-dimensional lattice with diffusing, cross-fed resources (Materials and methods; ODD description in S1 Text). Each genome is a gene regulatory network, specified by a weight matrix and node biases, with 13 to 32 nodes. Thirteen nodes are fixed. Twelve are read out as phenotypic traits (resource-uptake allocation, body size, motility, division threshold, DNA repair, stress tolerance and leakage) and one receives an abiotic cue. The remaining hidden nodes are added by duplication and removed by deletion, so genome size can evolve. The network is developed once at birth, and the phenotype does not change afterwards. Mutation can create and remove connections and each non-zero connection has a small maintenance cost, so connectivity is also free to evolve.

R-Aevol is the version of the Aevol platform that includes gene regulation [19–21]. Genomes are circular sequences in which promoters, genes and binding sites are identified by motif search. Translated proteins contribute to a phenotype, a function that is compared with a target function, and can also bind upstream of promoters to activate or repress the transcription of other genes. Protein concentrations are integrated over the 20 ages of an individual's life, and fitness is evaluated against the target at each age. Environmental signals are signal proteins that are present at a fixed concentration during the ages in which their environment is active, so organisms can evolve to respond to them.

The design (Fig 1) crosses environment (frozen or fluctuating) with cue (informative, absent or yoked). In R-Aevol the fluctuating environment switches between two target functions, A and B. B is the mirror image of A and has the same geometric area, so neither target is harder to match. In the frozen environment the switching mechanism is kept but the two targets are identical. In the informative arm each target is paired with its own signal protein. The yoked arm has four environments, one for each combination of target and signal, so the two signals occur equally often but say nothing about the target. At each switch R-Aevol moves to one of the other environments at random. At a given switching probability the four-environment arm thus changes its target less often than the two-environment arm, and we multiplied the yoked arm's switching probability by 1.5 so that the two arms have the same per-age rates of target change and of signal change (Materials and methods). We write q for the switching probability per age. EvoSim follows the same logic with an abiotic cue, whose yoked version is decoupled from the environmental state by running on an incommensurate period.

In EvoSim, fluctuation increases connection number but not wiring strength

Earlier EvoSim experiments had found that fluctuation increases connection density (S4 Text). We repeated the comparison with 180 populations (six arms, 30 seeds each, 150,000 time steps) and measured the strength of connections as well as their number. Without a cue, fluctuation raised count density, the proportion of possible connections that are non-zero, from 0.467 to 0.587 (+25.5%, $p < 10^{-4}$, $d = 2.2$; 28 surviving fluctuating and 30 frozen populations; Fig 2A). Strength density, the summed absolute weight divided by the number of possible connections, fell in the same networks from 0.207 to 0.184 (−10.9%, $p < 10^{-4}$, $d = -1.4$; Fig 2B). Total wiring strength did not change (38.0 in frozen and 37.5 in fluctuating populations, $p = 0.86$; Fig 2C). Fluctuating networks thus had more connections but not more wiring, and the average non-zero connection was about 30% weaker. Results with the informative and the yoked cue were similar. Node count stayed close to its minimum of 13 in this regime and increased only slightly (+0.6 nodes, $p = 0.047$).

Population size is a possible confound. Fluctuating populations were 40% smaller than frozen ones, as expected since in frozen dishes all inflow feeds the best-paying channel (S1 Text), and population size can change genome architecture by itself. Because the population sizes of the two arms did not overlap, the two effects cannot be separated statistically. Within each arm, however, larger populations had lower strength density (r between −0.31 and −0.47; Fig 2D). If population size were the cause, the smaller fluctuating populations should have had higher strength density, which is the reverse of what we found. The cue was also not used in this regime, and strength density was the same with informative and yoked cues ($p = 0.99$).

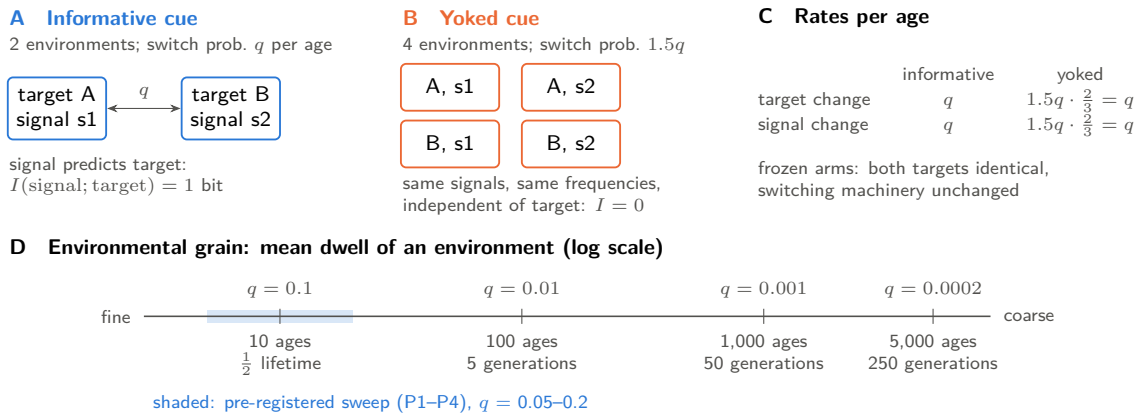


Fig 1. Design. (A) The informative arm in R-Aevol has two environments, each pairing a target with its own signal protein, so the signal predicts the target. (B) The yoked arm has four environments pairing each target with each signal, so the same signals occur at the same frequencies but carry no information about the target. (C) Because a switch moves to a different environment chosen at random, the yoked arm switches with probability $1.5q$, which gives it the same per-age rates of target change and signal change as the informative arm. Frozen arms use two identical targets. (D) Environmental grain as the mean dwell time of an environment, set by the switching probability q over the 20 ages of a lifetime. Ticks mark the four switching probabilities of the grain experiment (G1–G4); shading marks the range of the pre-registered sweep (P1–P4).

EvoSim organisms respond to the cue only when development is reversible, and even then they use a small part of the information it carries (S4 Text). The EvoSim results concern fluctuation that the organisms cannot track.

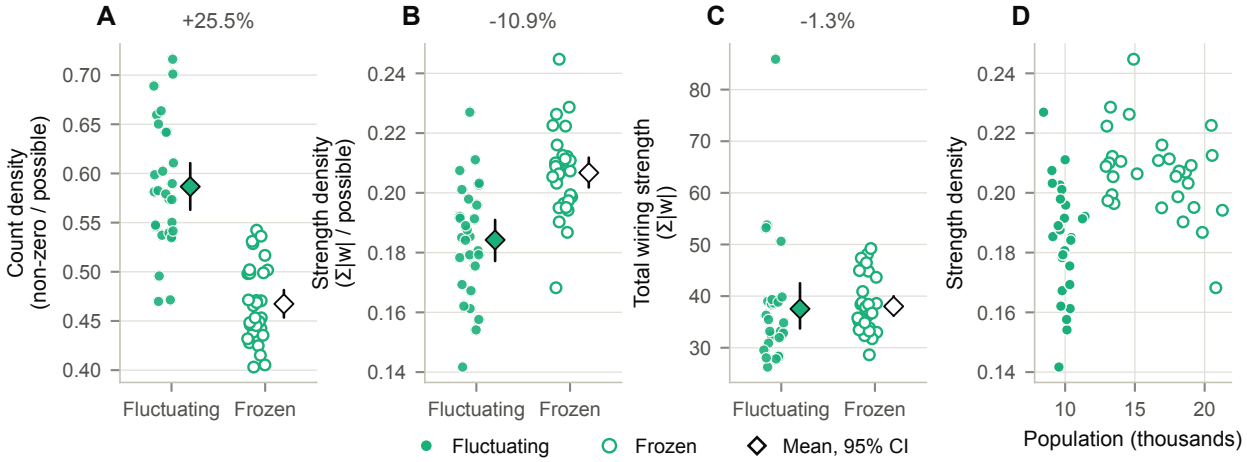


Fig 2. EvoSim: more connections, less strength. Fluctuating (no cue; filled) versus frozen (hollow) populations, one point per population; diamonds are means with bootstrap 95% confidence intervals. Numbers above panels are the change in the mean. (A) Count density, the number of non-zero connections divided by the number of possible connections. (B) Strength density, the summed absolute weight divided by the number of possible connections. (C) Total wiring strength. (D) Strength density against population size within each arm. Extinct populations (2 of 30 fluctuating, 0 of 30 frozen) are excluded.

In R-Aevol, the effect of fluctuation on regulation depends on the cue

We then asked whether an independent model would reproduce two earlier EvoSim conclusions, namely that fluctuation increases genome size and connectivity and that the information content of a cue makes no difference. It reproduced neither. In an exploratory experiment (six arms, 12 seeds each, 30,000 generations), fluctuation reduced genome size under every cue condition, and its effect on regulation depended on the cue (Fig 3). With an informative cue, total regulatory strength was 12.7 in fluctuating populations and 4.7 in frozen populations given the same signals. Fluctuating populations without a cue had the lowest regulatory strength of the six arms (1.4).

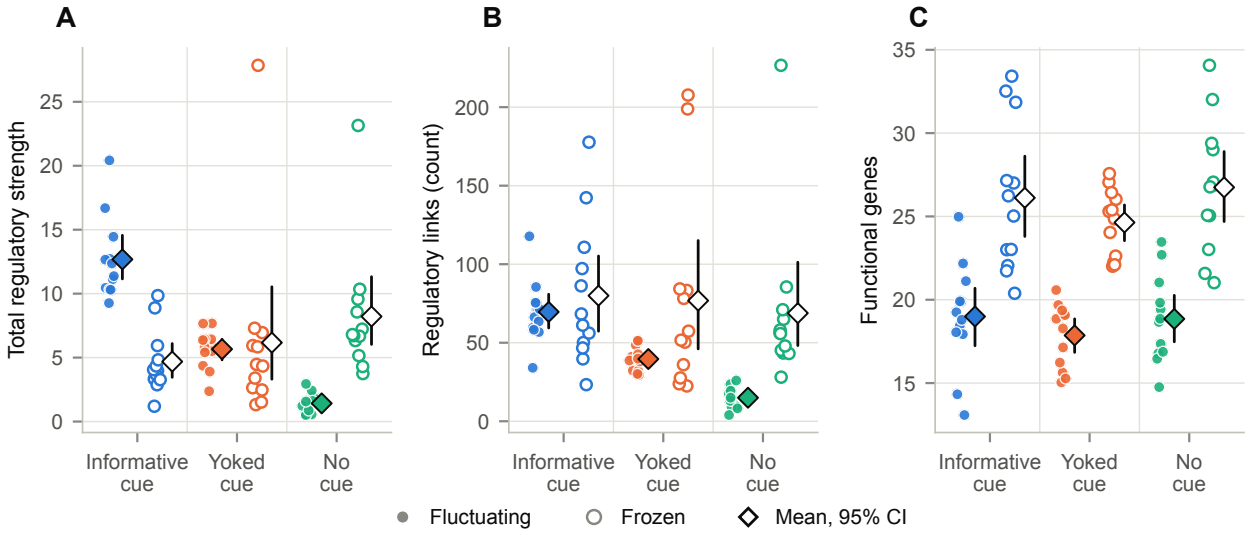


Fig 3. R-Aevol, six arms (exploratory). Total regulatory strength (A), number of regulatory links (B) and number of functional genes (C) for each combination of environment (filled, fluctuating; hollow, frozen) and cue (colour), one point per population; means with bootstrap 95% confidence intervals. $q = 0.10$, population 1024, 12 populations per arm.

Since this interpretation came from inspecting the data, we tested it with new seeds. Four predictions, with their measures, statistical tests and decision rules, were committed to version control before any of the new simulations were run (Table 2; Materials and methods). The switching probability took three values over a fourfold range ($q = 0.05$, 0.10 and 0.20) at a population size of 1024, with 12 new seeds per condition, and the main contrasts were repeated at a population size of 512. For the two predictions that depend on cue use we also specified a manipulation check. At a given rate, the informative cue had to reduce metabolic error relative to the yoked cue, showing that the organisms used it. The check was met at $q = 0.10$ and $q = 0.20$ ($p < 10^{-4}$) but not at $q = 0.05$ ($p = 0.088$), so under the pre-registered rule these two predictions were evaluated at the two faster rates.

All four predictions were supported (Table 2, Fig 4). With an informative cue, fluctuation increased total regulatory strength relative to the frozen environment 1.8-fold at $q = 0.10$ and 2.7-fold at $q = 0.20$ (P1; Holm-adjusted $p = 0.005$ and $p \approx 10^{-4}$). Without a cue, fluctuation lowered regulatory strength about fivefold at all three rates (1.1 to 1.2 compared with 6.3; P3; Holm-adjusted $p \approx 10^{-4}$ at each rate). The informative-cue cells at the slowest rate, which were excluded from the verdict, showed the same direction (11.0 compared with 5.9). In one of those populations every regulatory connection was lost for the whole analysis window, and no other population in the sweep lost them completely.

The information carried by the cue produces the effect

The yoked control separates information from stimulation. Informative and yoked arms receive the same two signal proteins at the same concentration and frequency, with the same switching rates, and differ only in whether the signal predicts the target. Informative-cue populations evolved 2.3 times ($q = 0.10$) and 2.9

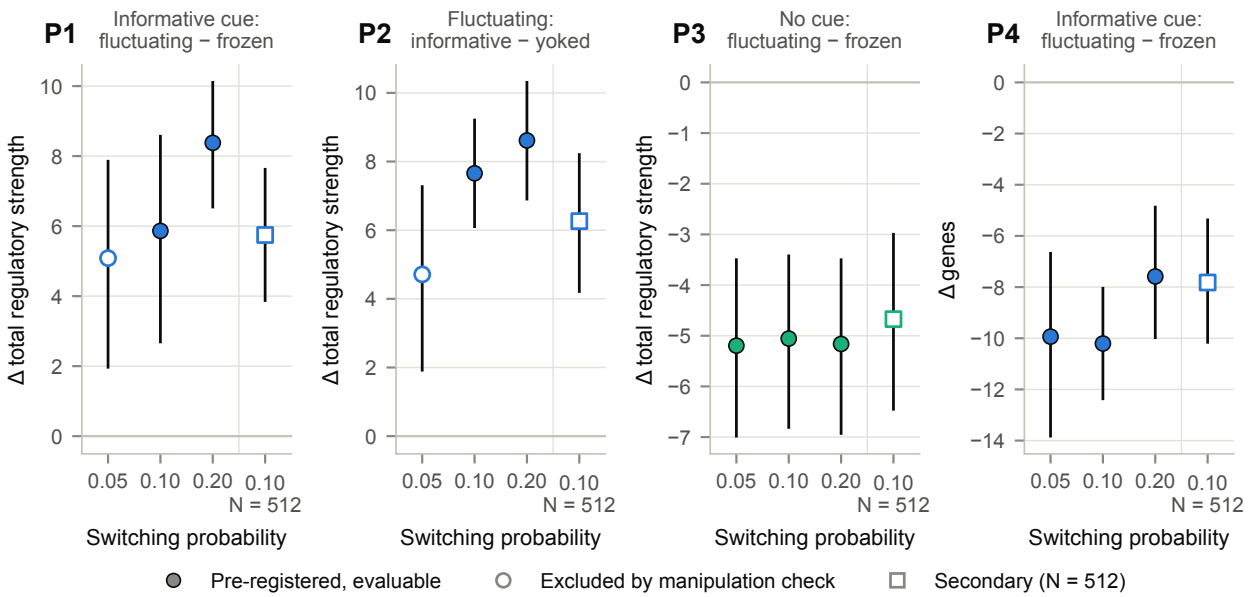


Fig 4. Pre-registered test. Differences in means with bootstrap 95% confidence intervals for each pre-registered contrast, at each switching probability (population 1024) and at population 512 (squares; secondary, no verdict). Open circles mark cells excluded from the verdict because the manipulation check failed at that rate. For P3 at population 512 the frozen baseline is the population-1024 frozen arm, the only one run. P1: effect of fluctuation on total regulatory strength with an informative cue. P2: informative minus yoked cue under fluctuation. P3: effect of fluctuation without a cue. P4: effect of fluctuation on the number of functional genes.

times ($q = 0.20$) the regulatory strength of yoked-cue populations (P2; Holm-adjusted $p \approx 10^{-4}$ in each case) and matched the target about twice as closely (metabolic error 0.029 against 0.058 at $q = 0.10$). In the exploratory experiment they also carried more transcription factors (11.2 against 6.2). In the frozen environment the signal still switches but predicts nothing, because the target does not change, and there the informative and yoked cues did not differ (7.5 and 5.4, $p = 0.18$). The difference between cue arms appears only when the cue carries information about a changing environment.

This result contradicts the earlier EvoSim conclusion that the information content of a cue has no effect. The two can be reconciled by how much the cue was used. In the EvoSim regime from which that conclusion came, organisms used only about a tenth of the available information (S4 Text), whereas in R-Aevol an informative cue halved metabolic error.

Fluctuation reduces genome size

Fluctuation reduced the number of functional genes at every switching rate, by 10, 10 and 8 genes (26.3, 28.3 and 25.6 in frozen populations compared with 16.3, 18.1 and 18.0 in fluctuating ones; P4; Holm-adjusted $p \leq 3 \times 10^{-4}$; Fig 4). The reduction was also present at a population size of 512 (25.3 compared with 17.5, $p < 10^{-4}$). Other models have reported the opposite, larger genomes under fluctuating selection [10, 11], and so did earlier EvoSim experiments (S4 Text). A possible explanation lies in how Aevol builds phenotypes. The phenotype is the sum of contributions from individual genes, so a constant target that can be matched precisely favours additional genes. A target that changes within a lifetime cannot be matched precisely, and the coarser phenotype it leaves needs fewer genes. Consistent with this, frozen populations reached a metabolic error of about 0.008 and fluctuating populations without a usable cue about 0.06. We have not tested this explanation directly, and the grain experiment described below does not support it strongly. Coarse-grained populations matched their targets better than fine-grained ones (metabolic error 0.038 against 0.062) but did not carry more genes.

Counting connections can give the wrong sign

Connection number and connection strength did not always change together (Fig 5). In EvoSim, fluctuation increased the number of connections and decreased strength per possible connection. In R-Aevol, fluctuating populations with an informative cue had fewer regulatory links than the corresponding frozen populations at every switching rate (0.60, 0.78 and 0.83 times as many), although their total regulatory strength was 1.8 to 2.7 times higher and their strength per link about twice as high. A study measuring connectivity as link number would have concluded that fluctuation with a usable cue makes regulatory networks simpler. Without a cue, number and strength fell together to about a fifth of the frozen values, so the two measures are not always in conflict. They can, however, diverge enough for a count to give the wrong direction. These comparisons were pre-specified as secondary analyses without a formal verdict.

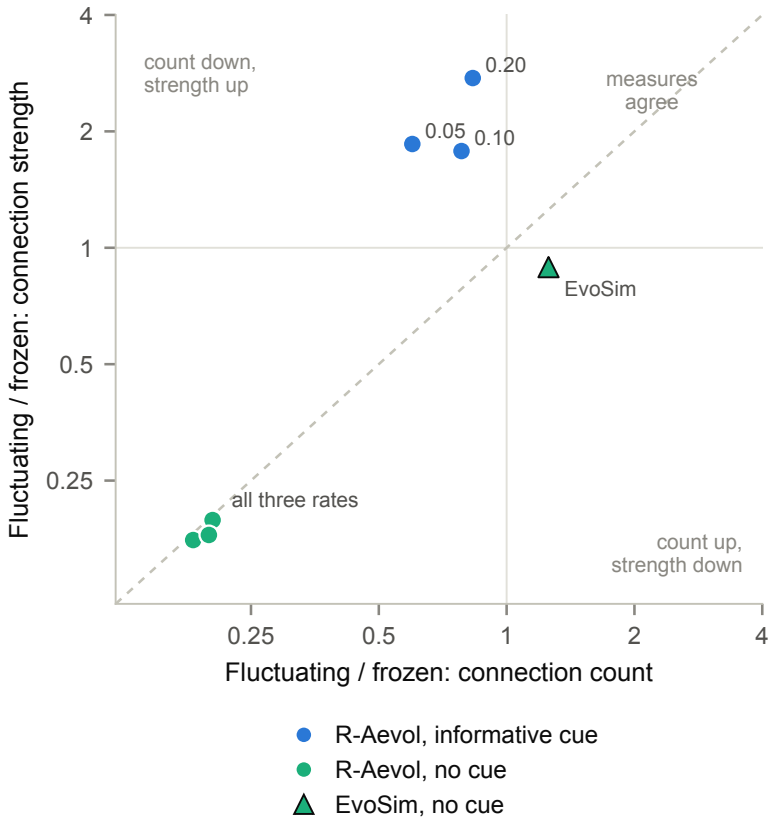


Fig 5. Counting versus weighing. Ratio of fluctuating to frozen connection count against the corresponding ratio of connection strength, on logarithmic axes. R-Aevol (circles): raw number of regulatory links and total regulatory strength, per switching probability, population 1024. EvoSim (triangle): count density and strength density, no cue. Points on the dashed diagonal are cases in which the two measures agree; points off it are cases in which counting misreports strength, and points in the upper-left or lower-right quadrants are cases in which it reports the wrong sign.

Environmental grain

Tsuda and Kawata [10] found that cue-free fluctuation across generations promotes regulatory complexity. To examine this disagreement we varied grain in R-Aevol without modifying the model. The environment is a single Markov chain over ages that continues from one generation to the next, so the switching probability sets how long an environment persists on average, from half a lifetime at $q = 0.1$ to about 250 generations at $q = 0.0002$ (Fig 1D). We predicted that coarse-grained cue-free fluctuation would produce

larger genomes and stronger regulation than fine-grained fluctuation (G1, G3) and than a frozen environment (G2, G4), the direction reported by Tsuda and Kawata. How each possible outcome would be interpreted was fixed in advance (Materials and methods).

The results did not support this prediction (Table 2, Fig 6). Under cue-free fluctuation, genome size and regulatory strength did not change with grain. At the coarsest grain, populations carried 19.2 genes and had a regulatory strength of 2.0, against 19.5 and 2.3 at the finest (G1 and G3, inconclusive; $p = 0.80$ and 0.60), and a secondary analysis found no trend across the four grains (Spearman correlation with log dwell time 0.06 for genes and 0.01 for strength). Coarse-grained fluctuation reduced gene number and regulatory strength relative to the frozen environment as much as fine-grained fluctuation did (25.6 genes and a strength of 8.4 in frozen populations; G2 and G4 refuted, Holm-adjusted $p = 4 \times 10^{-4}$ and 8×10^{-5}). Coarser grain did improve the match to the target, lowering metabolic error from 0.062 to 0.038, but genome size and regulation stayed where they were. The finest-grained cell reproduced P3 and P4 once more, with new seeds. By the interpretation fixed in advance, grain does not reconcile our results with those of Tsuda and Kawata.

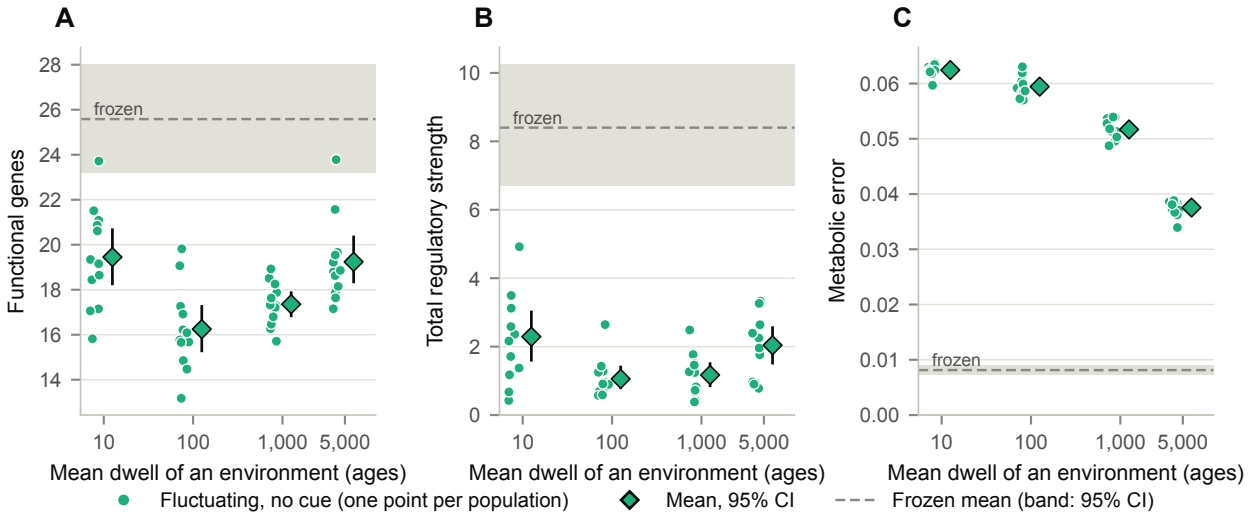


Fig 6. Environmental grain. Cue-free fluctuating populations at four switching probabilities, plotted against the mean time an environment persists ($1/q$ ages; 20 ages make one generation), one point per population, with means and bootstrap 95% confidence intervals. The dashed line and grey band show the mean and 95% confidence interval of the frozen arm. (A) Functional genes. (B) Total regulatory strength. (C) Metabolic error, the distance between phenotype and target.

Discussion

In R-Aevol, the effect of environmental fluctuation on regulation depended on whether organisms could identify the current environment. Fluctuation with an informative cue produced the strongest regulation we observed and fluctuation without a cue the weakest. A yoked cue, which supplied the same molecules at the same rates, did not have the effect of an informative one. These results held across switching rates and at two population sizes, and the cost of cue-free fluctuation held across a 500-fold range of environmental grain, in tests whose predictions and analyses were fixed before the data were generated.

The phenotypic side of this result is well known. Reliable cues favour plastic responses, and in their absence generalist phenotypes or bet-hedging are favoured [5–8]. This has recently been shown for the evolution of plasticity in a gene regulatory network [9]. Our results add the corresponding effect on the network that implements the response, in terms of its regulatory strength and its size. The yoked control also allowed us to separate the information carried by a cue from its physical presence, which cannot be done by varying cue reliability alone.

We designed the grain experiment to reconcile these results with those of Tsuda and Kawata, who found that cue-free fluctuation of the optimum across generations promotes regulatory complexity [10]. It did not. In R-Aevol, cue-free fluctuation reduced genome size and regulatory strength to the same extent whether an environment lasted half a lifetime or about 250 generations, and coarser grain improved how closely populations matched their targets without changing either quantity. Two differences between the studies remain as candidate explanations. Tsuda and Kawata's optimum moved to new positions in a random walk, whereas ours alternated between two fixed targets, and they found the random walk more favourable to complexity. A design with novel optima would test this directly. The other candidate is the way each model maps genotype to phenotype. The same applies to EvoSim, which produced larger genomes under fluctuation in earlier experiments. Its fluctuation is coarse relative to a lifetime, since development happens once at birth and each environmental state lasts about 14 generations, but the grain experiment makes coarse grain an unlikely explanation, and the difference in genome size between EvoSim and R-Aevol is most likely a difference between the models.

The divergence between connection number and connection strength has different causes in the two models. In EvoSim, gene duplication and node addition copy the connections of an existing node and add small random values to every copied weight, including weights that were zero, so each duplication creates many weak connections. When purifying selection is weak, as in the smaller populations of fluctuating environments, these connections persist, and connection number rises without an increase in total strength. In R-Aevol, frozen populations had more regulatory links, but they were weaker. Informative-cue populations had fewer links with roughly twice the strength. We did not determine why frozen populations accumulate weak links. Most evolutionary models of regulation measure complexity by counting connections [10, 14, 15], and we recommend that weighted measures be reported as well.

Limitations

The positive result, that a usable cue allows fluctuation to strengthen regulation, rests on R-Aevol alone. EvoSim organisms respond to a cue only when development is reversible and even then use little of it, and we have not measured regulatory strength in that regime. Each R-Aevol condition contained 12 populations, fewer than the 60 to 100 used in comparable studies [10, 22]. The effects were large ($|d|$ between 1.5 and 3.9 in the evaluated cells) and remained significant after correction for multiple testing, but smaller effects could have been missed. R-Aevol records statistics only for the best individual in each population. Its fluctuating environments alternate between two fixed targets and do not include the random walk towards new optima that Tsuda and Kawata found most favourable to complexity [10]. In EvoSim, fluctuation and population size are confounded between arms. The within-arm correlations argue against population size as the cause of the lower strength density, but they do not rule it out. Prediction P4 establishes the direction of the effect on genome size; its mechanism is untested.

We also pre-registered a confirmation of the EvoSim divergence between count and strength in a regime with more hidden nodes, on condition that fluctuating populations survived there. They did not (2 to 5 of 6 survived, compared with 6 of 6 frozen populations), so the confirmation was not run and the EvoSim result remains exploratory. Finally, all pre-registrations were committed to a version-control repository before the corresponding runs, but these timestamps exist only locally and cannot be verified independently.

Conclusion

Environmental fluctuation did not by itself strengthen gene regulation in these models. Regulation became stronger under fluctuation when a cue allowed organisms to track the environment, and weaker when no such cue existed, whatever the grain of the environment. Because connection counts and connection strength can change in opposite directions, conclusions about regulatory complexity may depend on which of the two is measured.

Materials and methods

EvoSim

EvoSim is described in full, following the ODD protocol [23], in S1 Text, and the parts relevant to this study are summarised below. Organisms live on a $G \times G$ lattice ($G = 128$) with six diffusing resource channels. External inflow, which varies seasonally, enters the first channel (with the exception described under Environment and cue), and consuming channel r releases a byproduct into channel $r + 1$, with yields declining along this chain. All arms also experience rare random mass-mortality and famine events, on average one every 25,000 time steps. Each organism has an energy store and divides when the store exceeds a heritable threshold. Starvation and the other causes of death are described in S1 Text.

Genome and development. A genome is a weight matrix W and a bias vector b over n_g nodes, with $13 \le n_g \le 32$. Nodes 1 to 12 are read out as the phenotype, node 13 receives the abiotic cue, and hidden nodes $14, \ldots, n_g$ have no fixed meaning. Development takes place once, at birth. Expression is initialised at 0.1 and updated eight times as

$$e_k \leftarrow \tanh\Big(b_k + \sum_m W_{mk}\, e_m + u_k\Big), \tag{1}$$

where u_k is the local concentration of resource k for $k \le 6$, the cue for $k = 13$, and zero otherwise. The final expression values are mapped to the twelve traits.

Mutation. At each birth, a Poisson number of point mutations with mean $10^{-3} n_g^2$ is applied, each adding $\mathcal{N}(0, 0.25^2)$ to a randomly chosen weight. This rate is reduced by up to 85% by the organism's heritable repair trait. The structural mutations are gene silencing (probability 0.002), which sets a node's outgoing weights to zero; gene duplication (0.002), which copies the regulatory role of one node onto another and adds $\mathcal{N}(0, 0.05^2)$ noise to every copied weight; and the addition or deletion of a hidden node (0.003 each), where an added node is a duplicate of an existing one made in the same way. Connection silencing (probability 0.2 per birth) sets one randomly chosen weight to exactly zero. Without it, additive mutation would drive density towards one. Each hidden node costs 0.0015 and each non-zero weight 5×10^{-5} energy units per time step.

Environment and cue. An abiotic condition cycles through discrete states with a period of 2,000 time steps. Each state changes which resource channel gives the highest yield, and in fluctuating arms 40% of the external inflow goes to the favoured channel. In frozen arms all inflow enters channel 1, which then also gives the highest yield. The condition, scaled to $[-1, 1]$, is the input to node 13. Fluctuating arms have six states and frozen arms one. In the yoked arm the cue is generated on a period incommensurate with the cycle (a multiplier of $5 + \varphi$), so that it covers the same range of values but carries only 0.0004 bits of information about the state. In the no-cue arm the gain of the cue is zero. With a single state the cue is identically zero, so the three frozen arms produce identical runs and were analysed as one arm.

Measures. Count density is the number of weights with $|w| > 10^{-4}$ divided by n_g^2. Strength density is $\sum |w| / n_g^2$ over all n_g^2 weights, including zeros, and total strength is $\sum |w|$. Each measure is computed for every organism at birth, averaged over the population at each logged time step, and then averaged over the last 20% of the run. Populations that went extinct were excluded and their number is reported.

Runs. We ran six arms with 30 seeds each for 150,000 time steps (180 runs), about 6,200 generations in fluctuating and 4,900 in frozen populations (24 and 31 time steps per generation, from the slope of mean lineage depth). Runs are deterministic for a given seed and bit-identical across thread counts.

R-Aevol

We used the `aevol-7` branch of Aevol (commit `33277374`, https://gitlab.inria.fr/aevol/aevol) compiled with regulation enabled (`-Dwith-raevol=ON`) and with per-generation statistics

(`-Dwithout-stats=OFF`). Building on Apple Silicon required correcting a hard-coded path to the OpenMP library (S2 Text). The parameters (Table 1) are taken from the example configurations distributed with the platform (`basic_regul`, `dynamic_env` and `complex_env`). The mutation and rearrangement rates are those of `complex_env`, ten times those of `dynamic_env`, and the binding matrix is the file supplied with `dynamic_env`, used unchanged in every run.

Environments. Each environment is a target function made of three Gaussians. Target B is target A reflected about the midpoint of the phenotypic axis, and the two have the same geometric area, 0.152468. Fluctuating arms switch between A and B. Frozen arms use two copies of the same target (A for odd seeds, B for even seeds), so that the switching mechanism runs in the same way but the organism never experiences a change. The environment is a Markov chain over ages that continues across generations. At each of the 20 ages it moves, with probability q, to a different environment chosen uniformly among the others. We checked in the source code that this is the switching carried out by the platform's SIMD engine (S2 Text).

Signals and the yoked control. Two signal proteins, taken unchanged from the `complex_env` example, are present at concentration 0.9 during the ages in which their environment is active and absent otherwise. The informative arm has two environments, (A, s1) and (B, s2). The yoked arm has four, (A, s1), (A, s2), (B, s1) and (B, s2). The no-cue arm has no signals. In the two-environment arm every switch changes the target and the signal together, so each changes at rate q per age. In the four-environment arm a switch changes the target with probability 2/3 and the signal with probability 2/3, so with a switching probability of $1.5q$ each again changes at rate q per age.

Measures. R-Aevol records statistics of the best individual in each generation. Genome size is the number of functional genes. Total regulatory strength is $S = n_{\text{link}}\,\bar{a}$, where n_{link} is the number of protein–RNA influences and $\bar{a}$ is their mean strength (the sum of the activation and repression factors divided by their number). $S = 0$ when there are no links. Each run is summarised by its mean over generations 25,001 to 30,000.

Runs. The exploratory experiment had six arms with 12 seeds each at $q = 0.10$ and a population of 1024 (72 runs). The pre-registered sweep comprised fluctuating informative, yoked and no-cue arms and frozen informative arms at $q = 0.05$, 0.10 and 0.20, together with frozen no-cue and frozen yoked arms at $q = 0.10$ (the frozen no-cue arm does not depend on q), each with 12 new seeds and a population of 1024 (168 runs). The fluctuating informative, yoked and no-cue arms and the frozen informative arm were repeated at a population of 512 (48 runs). The grain experiment has cue-free fluctuating arms at $q = 0.1$, 0.01, 0.001 and 0.0002 and a frozen arm, with 12 new seeds each (60 runs). All completed runs reached 30,000 generations.

Statistics

The unit of analysis is a population. Each run contributes one value per measure, so generations within a run are not treated as independent. Differences between arms were tested with two-sided permutation tests on the difference of means (100,000 permutations), and rank-based permutation tests are reported alongside. Multiple predictions were corrected with the Holm procedure. Effect sizes are given as Cohen's d, and intervals in the figures are bootstrap 95% confidence intervals (10,000 resamples).

Pre-registration and deviations

Predictions P1 to P4, their measures, tests and decision rules, and the analysis code were committed to version control (commit `f985ccd`) before the pre-registered sweep was started. The analysis was run once, on the complete data, with the committed code, whose checksum was verified beforehand. The only deviation was operational. The idle-sleep setting of the host machine suspended the runs overnight, which lengthened the wall time but cannot have affected the results, because runs are deterministic for a given seed. The EvoSim confirmation (commit `4bc16c3`) required fluctuating and frozen populations to survive a

Table 1. R-Aevol parameters.

Parameter	Value
Population	1024 (32×32) or 512 (32×16)
Selection	fitness-proportionate, pressure 750
Point mutation, small insertion, small deletion	5×10^{-5} per base per generation
Duplication, deletion, translocation, inversion	5×10^{-5} per base per generation
Maximum indel size	6
Initial genome	5,000 bases, one functional gene
Hill function	$n = 4$, $\theta = 0.5$
Degradation	rate 1, 10 steps per age
Lifetime	20 ages, fitness evaluated at all 20
Binding matrix	fixed file from `examples/dynamic_env`, all runs
Target sampling	300 points
Signal concentration	0.9
Generations	30,000

viability pilot. None of the three candidate regimes met this condition, so it was not run (commit `56fe3a0`). The design, predictions and analysis code of the grain experiment were committed in the same way (commit `fa00777`) before its runs began, and its analysis was run once, automatically, when the last run finished.

Table 2. Pre-registered predictions and verdicts.

Id	Prediction	Verdict
P1	Fluctuation with an informative cue raises total regulatory strength over frozen	supported ($q = 0.10$, 0.20)
P2	An informative cue raises it over a yoked cue	supported ($q = 0.10$, 0.20)
P3	Fluctuation without a cue lowers it below frozen	supported (all rates)
P4	Fluctuation lowers the number of functional genes	supported (all rates)
P5	In EvoSim, count density rises and strength density falls in a regime above the node floor	not run (no viable regime)
G1, G3	Coarse-grained cue-free fluctuation gives more genes (G1) and stronger regulation (G3) than fine-grained	inconclusive
G2, G4	Coarse-grained cue-free fluctuation gives more genes (G2) and stronger regulation (G4) than frozen	refuted

A prediction was supported if the predicted sign held at every evaluable rate and the Holm-adjusted $p < 0.05$ at two or more, refuted if the opposite sign was significant at any rate or no rate was Holm-significant, and inconclusive otherwise. P1 and P2 are evaluable only at rates where the informative cue measurably lowers metabolic error relative to the yoked cue. G1–G4 are single tests, supported if the predicted sign holds with Holm-adjusted $p < 0.05$, refuted if the opposite sign holds with $p < 0.05$, and inconclusive otherwise.

Software and availability

EvoSim is written in Julia and uses only the standard library for simulation. The R-Aevol runs were made on a single 12-core machine. The configurations of the two models, the launch scripts, the pre-registered analysis scripts, the history of all analyses and the per-run statistics are available at **[PENDING:** repository URL and Zenodo DOI].

Supporting information

S1 Text. ODD description of EvoSim. Complete model description following the ODD protocol, with every parameter value.

S2 Text. R-Aevol build, configuration and the yoked-control derivation. Build options and the changes needed on Apple Silicon, the parameter-file template, targets and signals, the environment-switching code, the derivation of matched switching rates, the definition of the measures, seeds, and the pre-registration record.

S3 Table. Results for every arm of every experiment. Means and standard deviations for all 528 runs. Per-run values are in the repository.

S4 Text. Earlier EvoSim results and the audit trail. Status of earlier claims after this study, the record of pre-registered predictions, and the measurement errors found during the project with the checks that exposed them.

S5 Table. Prior work and how this study relates to it.

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